

The Making Of A Sturdy And Secure AUTONOMOUS E-TWO-WHEELER

For a country like India, where more than half of the population is dependent on two-wheelers, enhancing these vehicles will be nothing short of achieving the dream of implementing autonomy on a wider scale



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Two-wheelers have been an integral part of the Indian lifestyle for many years. While the public can smoothly go about its business, tourists can easily traverse the length and breadth of this vast country with these vehicles. Even Bollywood has been hugely influenced by it, so much so that many key scenes have involved a two-wheeler in action.

The idea of developing an autonomous two-wheeler amongst the current excitement around autonomous cars may be seen as silly by some. But the fact is that the idea of having an autonomous e-two-wheeler has been there since the

early 2010s. So, is the idea of having an autonomous e-two-wheeler relevant more than a decade later? The answer is a full-fledged yes. If it could be thought of back then, it surely is possible now when technology has taken great strides.

The need

In America and Europe, self-driven cars are quite popular, and you see very few two-wheelers on the road. The two-wheelers are used there mainly for leisure and sports. In India, two-wheelers are the lifeline of our transportation system. They are much less expensive here, so more people buy them.

But maintaining a smooth traffic flow due to the huge variety of vehicles on the Indian roads is a challenge for the authorities. Building more lanes and flyovers helps, but for a country like India their construction remains slow due to monetary constraints.

Two-wheelers occupy lesser space compared to four-wheelers, so more people can utilise each lane. Thus, there is lesser requirement for new lanes and flyovers. Therefore, in the future, when the traffic is not likely to reduce, the situation can be better tackled by using two-wheelers instead of cars.

The technology

Standard autonomous mobile robot models, for which a lot of research has been done over the last several decades, include the unicycle model, the bicycle model, the differential wheel type model, the tricycle

Lane splitting

A common road phenomenon observed in the USA and the UK is 'lane splitting' wherein two-wheelers travel between adequately spaced rows of four-wheelers. And because two-wheelers are extremely agile, in case of a traffic jam, they can easily navigate their way ahead. So, for an efficient transportation system, lane splitting is the way to go in countries with a greater number of smaller vehicles.



Lane splitting at a traffic light in New York (Credit: Unsplash)